Particle model of solids, liquids and gases

1 For each of the substances below, show whether it is a solid, a liquid or a gas by joining the boxes

ice	**liquid**
oxygen	**solid**
petrol	**gas**

2 Complete these sentences by writing ‘‘shape’’ or ‘‘size’’ in each space.

a Solids keep the same _____ and _____.

b Liquids keep the same _____ but not the same _____.

c Gases do not keep the same _____ or _____.

3 Which of these statements explains why a can collapses when the air is removed from it.

a There are more air particles outside the can than inside.

b There are more air particles inside the can than outside.

c The air particles inside the can aren’t pushing as hard as they were.

d All the air has been sucked out.

4 Betty says that jelly is a liquid, but Al says that jelly is a solid.

a Write down one observation that supports Betty’s argument. _____

_____

b Write down one observation that supports Al’s argument. _____

_____

c Is jelly a solid , a liquid, a gas or a mixture? _____

_____

5 Geri did some experiments where she tried to squash solids, liquids and gases in syringes. She wrote down the following statements after her experiments. Which are true and which are false?

a Gases can be squashed true / false

b Liquids can be squashed true / false

c Solids cannot be squashed true / false

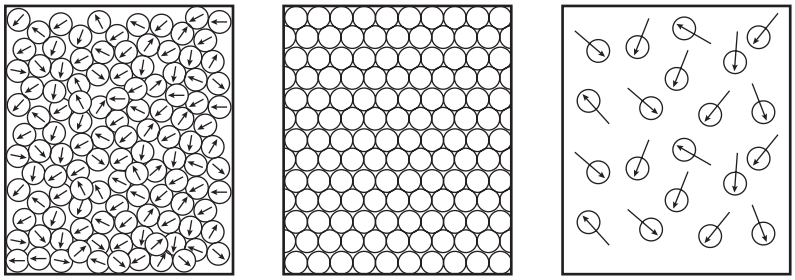
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Particle model of solids, liquids and gases continued

6 **a** Which diagram shows the arrangement of particles in a solid?

b Which diagram shows the arrangement of particles in a liquid?



A B C

c Which diagram shows the arrangement of particles in a gas?

7 Simon investigated particles by heating a bar and seeing how it fitted in a gauge. Before he heated the bar, it fitted perfectly in the gauge, but when he heated the bar it changed. Which of the following statements are true, and which are false?

a the bar expanded

b the bar contracted

c the bar got longer

d the bar got shorter

e the bar got thinner

f the bar got thicker

g the particles in the bar expanded

h the particles in the bar contracted

i the particles in the bar moved further apart

j the particles in the bar moved closer together

8 Join a box in the first column to a box in the second column and then join the second box to a box in the third column to make complete sentences.

a	flour is a mixture of	solid and	solid
	jelly is a mixture of	gas and	solid
	sponge is a mixture of	liquid and	gas
b	sand is a mixture of	gas and	liquid
	hair gel is a mixture of	liquid and	solid
	foam is a mixture of	solid and	gas

9 Choose the best statement from those below to explain why diffusion happens when gases mix.

a Both gases are made of particles.

b Particles are very small.

c Particles move around all the time.

d Gas particles are attracted towards each other.

10 Choose the best statement from those below to explain why gases have a pressure.

a Gases are made of particles which push against each other.

b Particles are very small and move around.

c Particles move and hit the walls of their container.

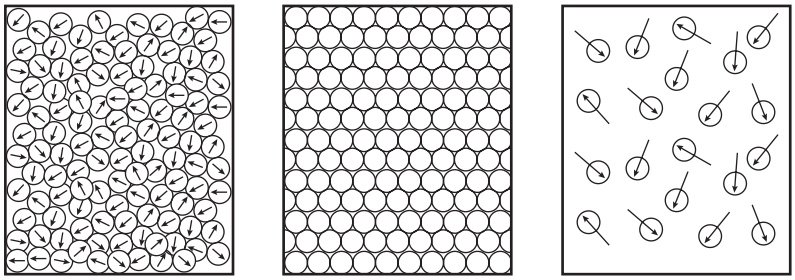
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